

Graduation Showcase: The Junior Innovator's Guide

Mastering Design Thinking & Functional Innovation

Welcome to the World of Design!

Did you know that you are already a designer? Every time you find a way to make something work better, you are using **Design Thinking**.

What is a Gadget? A gadget is more than just a piece of technology. It is a tool that solves a problem for people. It can be as simple as a tool to pump air or as complex as a computer.

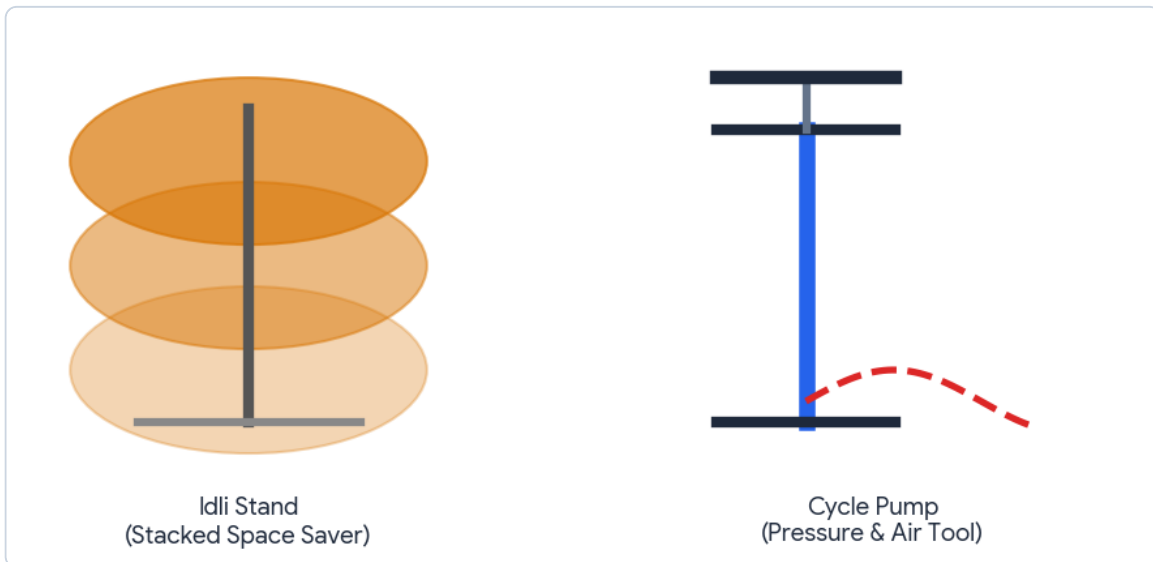
Your Mission Today: In this lesson, you will learn how real heroes use simple materials to solve big problems, and you will start your own journey using the **Design Loop**.

Mini-Task: Look around right now. Spot 3 gadgets you use every day. Think about your water bottle, cycle pump, or lunchbox.

1. Functional Design

Design is not just about how things look — it's about how they work! We call this **Functional Design**: design made to serve a special purpose.

- **The Idli Stand:** A brilliant design that saves space by stacking plates on top of each other.
- **The Cycle Pump:** A functional tool that keeps your bike ready to ride whenever you need it.



Visual Illustration 1: Functional Design Principles in Action

2. Meet Real-Life Heroes

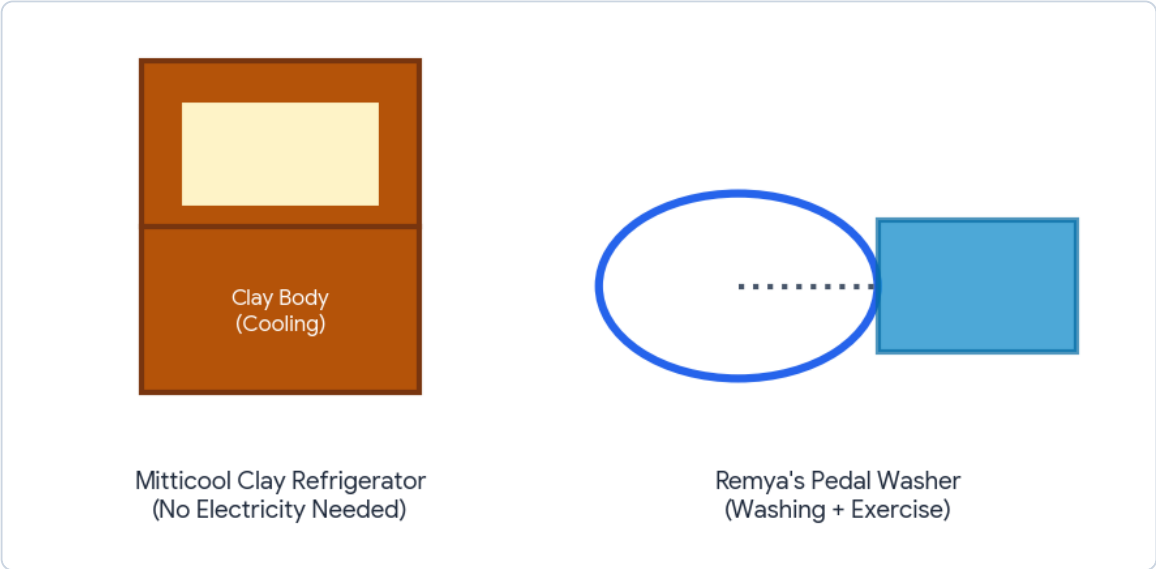
Meet two amazing Indian innovators who turned simple ideas into gadgets that changed lives:

Mansukhbhai Prajapati (The Clay Innovator)

He created *Mitticool*, a refrigerator made completely out of clay! It keeps water, milk, and vegetables fresh and cool without using any electricity.

Remya Jose (The Bicycle Laundry Star)

When she was a school student, she invented a washing machine attached to a bicycle! By pedalling the cycle, she could wash clothes while exercising, saving both electricity and time.



Visual Illustration 2: Real-Life Gadget Innovations

3. Student Showcase Ideas: What You Have Already Learned & Can Make!

Look at what students like you have already built and showcased using Design Thinking! You can use these ideas to inspire your own showcase project:

 Student Showcase Portfolio & Project Ideas

Category / Theme	What Students Do & Make (Showcase Ideas)
Home & Eco Gadgets	Create mini clay-based cooling boxes (inspired by Mitticool), zero-electricity food preservers, or simple eco-friendly water dispensers.
Mechanical & Motion Tools	Build pedal-powered or lever-operated mini machines (like pedal spinners, manual egg beaters, or rubber-band propelled sweepers).
Space-Saving Designs	Design multi-tiered vertical organizers, collapsible pen stands, or stackable lunch trays inspired by the Idli Stand.
School & Desk Solutions	Construct anti-roll pencil holders, spill-proof tiffin bag liners, or magnetic book-mark clips using household items.

4. The Design Loop: THINK

How do inventors come up with great ideas? They use a map called the **Design Loop**! The loop has three main steps that keep going around: **THINK, MAKE, and SHARE**.

STEP 1: THINK — Find the Problem

Every great gadget starts by looking around and finding a problem to solve. Design thinkers are always asking questions like:

- What is hard to do?
- What breaks often?
- How can we make this task more fun or faster?

Brainstorming Challenge:

Look at this list of daily hassles. Which one would you like to build a gadget for?

- The water bottle is too heavy to carry.
- Pencils keep rolling off the desk and losing their tips.
- The tiffin box leaks oil into the school bag.

5. The Design Loop: MAKE

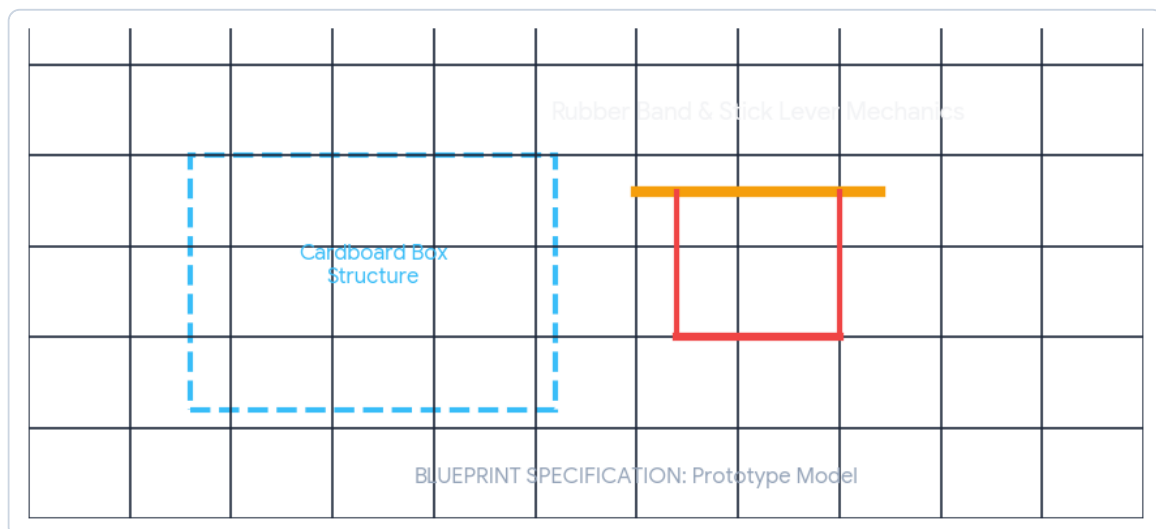
STEP 2: MAKE — Build and Test

This is where you become a maker! You use simple materials to create a **Prototype** — a rough model of your gadget.

The Materials Station:

You don't need expensive factory tools. You can make an amazing prototype using things found around your house or classroom:

- ✓ Cardboard boxes
- ✓ Plastic cups & bottles
- ✓ Rubber bands
- ✓ Popsicle sticks
- ✓ Straws
- ✓ Paper clips
- ✓ String & Tape
- ✓ Clay



Visual Illustration 3: Blueprint Drawing & Model Construction Setup

6. The Design Loop: SHARE

STEP 3: SHARE — Tell Your Story

Show your gadget to your friends, family, and classmates. Tell them how it helps solve the problem you found.

Why Sharing Matters:

- **Get Better Ideas:** When others see your gadget, they might have great suggestions to improve it.
- **Inspire Others:** Your invention could inspire someone else to create something amazing too!

Sharing Challenge:

Prepare a "30-second pitch" for your gadget. Explain:

1. What the problem was.
2. How your gadget solves it.
3. One thing you are proud of!

7. 3-2-1 Exit Review & Final Challenge

Let's check what we've learned today to make sure you're ready to be an inventor!

- **3 Gadgets:** List 3 gadgets you learned about today (from the lesson or your surroundings).
- **2 Steps:** Write down 2 steps of the Design Loop that you remember.
- **1 Feeling Word:** What is one word that describes how you feel about being an innovator?



Your Final Graduation Challenge!

Find one problem at home or school. Use the Design Loop to **THINK**, **MAKE**, and **SHARE** your gadget! Bring your idea or prototype to the graduation showcase.

Practice Worksheet: Future Innovators Challenge

Show what you know about Design Thinking! Answer the questions below using what you learned.

Question 1: What is a gadget? Give one simple example mentioned in the text.

Question 2: Explain what Functional Design means in your own words.

Question 3: How does the Idli Stand show good functional design? What problem does it solve?

Question 4: What amazing clay gadget did Mansukhbhai Prajapati create, and why is it special?

Question 5: How did Remya Jose find a way to wash clothes and exercise at the same time?

Question 6: Name the three main steps of the Design Loop.

Question 7: What questions do design thinkers ask when they are in the THINK stage?

Question 8: What is a Prototype, and during which step of the Design Loop do you build it?

Question 9: List four simple everyday materials you can use to build a prototype at home.

Question 10: Why is it important to SHARE your invention with friends and family? Give two reasons.

Question 11: Imagine your pencil keeps rolling off your desk. Write down one quick idea to fix this problem.

Question 12: What are the three things you should explain when preparing a 30-second pitch?